

How to Write a Research Paper

A practical guide to structuring, supporting, and presenting academic research

Prepared as a concise handbook for students and early-stage researchers

This guide explains the major sections of a standard research paper and shows what each section must accomplish. It emphasizes logical flow, evidence-based writing, and alignment among the research gap, objectives, questions, hypotheses, methods, results, and conclusions.

Recommended Writing Logic

1. Start from a meaningful problem, not just an interesting topic.
2. Use the literature to show what is known, what remains unresolved, and why the unresolved part matters.
3. Translate that gap into clear objectives, research questions, and where appropriate, hypotheses.
4. Choose data and methods that directly answer the questions.
5. Report results objectively before interpreting them.
6. Explain what the findings contribute to theory, practice, and future research.

1. Introduction

The introduction frames the study, explains why the topic matters, identifies the research gap, and shows how the paper will address it. A strong introduction moves from the broad problem to the specific unresolved issue and ends with a clear statement of purpose. It should persuade the reader that the study is timely, relevant, and necessary (Creswell & Creswell, 2018; Locke et al., 2022).

What the introduction should include

- The practical, social, managerial, or scientific importance of the topic.
- A brief review of what prior studies have already established.
- A precise statement of the gap, inconsistency, limitation, or unanswered question in the literature.
- The purpose of the current study and its expected contribution.
- The research questions and, where relevant, hypotheses.

Gap identification

A research gap is not simply a statement that few studies exist. A stronger gap explains what is missing and why that missing element matters. Common forms of gaps include theoretical gaps,

methodological gaps, contextual gaps, inconsistent findings, population gaps, temporal gaps, and construct gaps (Booth et al., 2016; Hart, 2018).

- Theoretical gap: prior studies explain the phenomenon with limited or competing theories.
- Methodological gap: previous work uses designs that cannot fully address causality, mechanisms, or context.
- Contextual gap: existing evidence comes from industries, countries, or populations different from the present study.
- Empirical inconsistency: earlier findings conflict and require clarification.

Research objective illustration

After the gap is stated, the paper should show how the study addresses it. Research objectives should be specific, observable, and aligned with the method. A useful formula is: “The objective of this study is to examine, explain, compare, test, or develop X in order to address Y.”

Objectives may be descriptive, explanatory, predictive, comparative, or theory-building.

- Weak objective: To study social media and performance.
- Stronger objective: To examine how social media analytics capability influences marketing decision quality in small firms.

Research questions with support

Research questions should emerge from the identified gap and the logic of the literature. Each question should be answerable using the planned data and methods. Questions should be neither so broad that they become unmanageable nor so narrow that they add little value. Good questions are clear, focused, and analytically actionable (White, 2017).

- RQ1 example: What factors explain variation in user adoption of the focal technology?
- RQ2 example: How does the proposed theoretical mechanism operate under different contextual conditions?

2. Literature Review

The literature review synthesizes prior work relevant to the research problem. Its purpose is not to summarize article after article in isolation, but to organize prior knowledge, evaluate the strength of evidence, identify patterns, and establish the need for the study (Booth et al., 2016; Snyder, 2019).

What a good literature review does

- Defines key concepts and clarifies how the study uses them.
- Groups prior studies by themes, constructs, debates, methods, or contexts.
- Shows agreements, disagreements, and unresolved issues.
- Builds the platform for the theoretical background and hypotheses.

Recommended writing approach

Organize the literature by logic, not by author chronology alone. For example, review prior work under thematic subheadings such as antecedents, mechanisms, outcomes, moderators, and boundary conditions. Within each subsection, compare studies rather than describing them one by one. A synthesis-oriented review demonstrates scholarly control and makes the later argument more persuasive.

- Avoid a “laundry list” review in which each paragraph begins with an author name.
- End each major subsection by explaining what remains unclear and how that informs the current study.

3. Theoretical Background (Theory Lens)

The theoretical background explains the conceptual lens used to interpret the phenomenon. A theory lens is important because it provides the logic that connects constructs, predicts relationships, and justifies why particular variables matter (Corley & Gioia, 2011; Whetten, 1989).

How to use a theory lens

- State the theory clearly and define its core assumptions.
- Explain why the theory is appropriate for the research problem.
- Translate abstract theoretical ideas into the constructs used in the study.
- Show how the theory helps derive the hypotheses or propositions.

The theory section should do more than name a theory. It should demonstrate the fit between the theory and the study context. If multiple theories are used, clarify the role of each and avoid unnecessary conceptual overlap.

4. Hypotheses Development and Research Model

Hypotheses development turns the reviewed literature and theory into testable expectations. Each hypothesis should be supported by a logical argument grounded in prior evidence and theoretical reasoning (Hair et al., 2022). In conceptual papers or qualitative studies, this section may instead develop propositions or a conceptual framework.

How to write a hypothesis paragraph

7. Define the focal constructs.
 8. Explain the causal or associative mechanism linking them.
 9. Support the mechanism with prior studies and theory.
 10. State the hypothesis in a direct sentence at the end of the paragraph or subsection.
- Example form: Based on theory X and evidence from prior studies, we expect that A is positively associated with B.

- Moderation form: The positive relationship between A and B will be stronger when C is high.
- Mediation form: A influences B through M because the theory suggests that A first changes M, which then shapes B.

Research model

The research model visually and verbally summarizes the proposed relationships among constructs. Even if a diagram is included, the text must still explain what arrows mean, which variables are dependent or independent, and where moderators, mediators, or controls fit. The model should match the hypotheses exactly.

Checklist for the hypotheses and model section

Element	What to verify
Construct definitions	Each variable is clearly defined and consistent with cited literature.
Logical support	The argument explains why the relationship should exist, not only that prior work mentioned it.
Directional clarity	The hypothesis clearly states positive, negative, mediated, moderated, or non-directional expectations.
Model alignment	The visual model matches the wording and numbering of hypotheses.
Boundary conditions	Moderators and contextual assumptions are explicitly justified.

5. Data Collection

This section explains where the data came from, who or what was studied, how participants or cases were selected, when data were gathered, and what procedures ensured quality and ethics. The description must be detailed enough for the reader to judge rigor and replication potential (Yin, 2018).

- State the source of data: survey, experiment, interviews, archival records, observations, panels, secondary databases, or mixed sources.
- Describe the target population and sampling strategy.
- Report the sample size and response or retention rate where relevant.
- Clarify the period of data collection and any screening or exclusion criteria.
- Mention ethical approval, informed consent, confidentiality, or anonymization when applicable.

6. Research Methodology

The methodology section explains how the study was designed and how the data were analyzed. It should allow readers to understand whether the method is appropriate for answering the research questions. Common components include research design, measures, reliability and validity checks, analytical techniques, and control variables (Field, 2018; Hair et al., 2022).

Typical subsections

- Research design: cross-sectional, longitudinal, experimental, quasi-experimental, case study, grounded theory, panel, or mixed-method.
- Measures and instrumentation: item sources, scale anchors, coding rules, or operationalization procedures.
- Data preparation: missing data handling, outlier treatment, transformations, and assumptions checks.
- Analysis plan: regression, structural equation modeling, qualitative coding, thematic analysis, difference-in-differences, content analysis, or other methods.
- Validity and reliability: construct validity, internal consistency, interrater reliability, robustness checks, triangulation, or sensitivity analysis.

Methodological choices must be justified. For example, if the study uses regression, explain why regression fits the question and data structure. If the study uses interviews, explain the logic of case selection, coding, and trustworthiness procedures.

7. Results

The results section reports what the analysis found. It should be factual, orderly, and directly tied to the research questions and hypotheses. Interpretation should begin only after the results are clearly presented. For quantitative work, report descriptive statistics, tests of assumptions where relevant, main model results, robustness checks, and whether each hypothesis is supported. For qualitative work, report themes, categories, representative evidence, and the analytic pattern linking findings to the questions (Miles et al., 2020).

- Present tables and figures that help the reader see the findings efficiently.
- Use consistent numbering and refer to each table or figure in the text.
- Report effect sizes, confidence intervals, and significance levels where appropriate, not only p-values.
- Keep interpretation modest in this section. Save broader meaning for the discussion.

8. Discussion

The discussion interprets the findings and explains what they mean relative to the research questions, theory, and prior literature. This section answers the “so what?” question. It should

explain not only what was found, but why the findings matter and how they change understanding of the focal phenomenon (Bem, 2004).

Core moves in a strong discussion

- Restate the central findings without repeating the results section word for word.
- Explain how the findings support, extend, refine, or contradict prior research.
- Interpret surprising, null, or mixed findings thoughtfully.
- Return to the theoretical lens and explain what the study reveals about it.

9. Theoretical and Practical Contribution

Many papers separate contributions into theoretical and practical parts. Theoretical contribution explains how the study advances academic understanding. Practical contribution explains how the findings can inform decisions, policies, design, or managerial action. Contributions should be specific and traceable to the actual findings, not exaggerated claims (Corley & Gioia, 2011).

Examples of theoretical contribution

- Introducing a new mechanism that links two established constructs.
- Clarifying boundary conditions that explain inconsistent prior results.
- Extending an existing theory to a new context or level of analysis.

Examples of practical contribution

- Providing evidence that helps managers allocate resources more effectively.
- Offering design guidance for systems, policies, or interventions.
- Identifying risk factors or leverage points practitioners can monitor.

10. Limitations and Future Research

No study is perfect. A good limitations section shows intellectual honesty and helps readers interpret the scope of the findings. Limitations should be substantive, relevant, and linked to future research opportunities. Avoid defensive writing and avoid listing trivial weaknesses (Creswell & Creswell, 2018).

- Typical limitations include sample constraints, measurement limitations, research design constraints, common method concerns, context dependence, and limited generalizability.
- Future research should logically follow from the study's boundary conditions, unresolved mechanisms, alternative contexts, or methodological improvements.

11. Conclusion

The conclusion closes the paper by briefly restating the research problem, the approach, the key findings, and the main contribution. It should leave the reader with a clear sense of why the

study matters. A conclusion is not a place for long new arguments. It is a concise synthesis that reinforces the value of the research.

12. Final Writing Tips

- Keep alignment across sections: the gap should lead to the objective, the objective to the questions, the questions to the method, and the method to the results.
- Use signposting sentences so readers can follow the logic of the paper.
- Revise for clarity, precision, and paragraph flow after the technical content is complete.
- Check journal or instructor requirements for section names, word limits, tables, appendices, and formatting.
- Use citation management software and verify every in-text citation against the reference list.

References

- Bem, D. J. (2004). Writing the empirical journal article. In J. M. Darley, M. P. Zanna, & H. L. Roediger III (Eds.), *The compleat academic: A practical guide for the beginning social scientist* (2nd ed., pp. 185-219). American Psychological Association.
- Booth, W. C., Colomb, G. G., Williams, J. M., Bizup, J., & Fitzgerald, W. T. (2016). *The craft of research* (4th ed.). University of Chicago Press.
- Corley, K. G., & Gioia, D. A. (2011). Building theory about theory building: What constitutes a theoretical contribution? *Academy of Management Review*, 36(1), 12-32.
- Creswell, J. W., & Creswell, J. D. (2018). *Research design: Qualitative, quantitative, and mixed methods approaches* (5th ed.). SAGE.
- Field, A. (2018). *Discovering statistics using IBM SPSS Statistics* (5th ed.). SAGE.
- Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2022). *Multivariate data analysis* (8th ed.). Cengage.
- Hart, C. (2018). *Doing a literature review: Releasing the research imagination* (2nd ed.). SAGE.
- Locke, L. F., Silverman, S. J., & Spirduso, W. W. (2022). *Proposals that work: A guide for planning dissertations and grant proposals* (7th ed.). SAGE.
- Miles, M. B., Huberman, A. M., & Saldana, J. (2020). *Qualitative data analysis: A methods sourcebook* (4th ed.). SAGE.
- Snyder, H. (2019). Literature review as a research methodology: An overview and guidelines. *Journal of Business Research*, 104, 333-339.
- Whetten, D. A. (1989). What constitutes a theoretical contribution? *Academy of Management Review*, 14(4), 490-495.

White, P. (2017). *Developing research questions: A guide for social scientists*. Palgrave Macmillan.

Yin, R. K. (2018). *Case study research and applications: Design and methods* (6th ed.). SAGE.